

Here are all the questions from the "INTERNET TECHNOLOGY" examination paper:

Group A [CO1: Total Marks: 10 (=5x2)]

Answer Any 5 questions from Q1:

1. Answer any five questions from the following:
 - a) How was the idea of Internet formed?
 - b) What is the role of the Internet Service Providers (ISPs)?
 - c) What speeds are in offer by the ISPs?
 - d) What are the core components of the Internet infrastructure?
 - e) What is meant by protocol?
 - f) How did TCP and IP evolve?
 - g) What is meant by 5G?

Group B [CO2: Total Marks: 40 (=10x1+1x30)]

Question no. 2 is COMPULSORY [Marks: 10]

Answer Question No. 3 OR Question No. 4 [Marks: 30]

2. Pick the correct answer:

- a) Mobile IP registration request contains:
 - i) Home agent and COA
 - ii) home address and home agent
 - iii) home address, home agent and COA
- b) In Snooping TCP, Foreign Agent (FA) detects loss of segment by looking into the:
 - i) sequence number
 - ii) timestamp
 - iii) NACK
 - iv) None of these
- c) Correspondent Node is:
 - i) communication partner
 - ii) home agent
 - iii) foreign agent
- d) First two bytes of Class B address are in the range of:
 - i) 128-191
 - ii) 128-190
 - iii) 191-223
 - iv) 192-223
- e) Who sends the acknowledgement to the Correspondent Node in I-TCP?
 - i) Mobile Host
 - ii) Access Point
- f) IPv6 addresses can be:
 - i) unicast, multicast and anycast
 - ii) only unicast
 - iii) only anycast
 - iv) only multicast
- g) Address length of IPv4 is:
 - i) 16 bits
 - ii) 32 bits
 - iii) 64 bits
 - iv) 128 bits
- h) The address [11000001 10000011 00011011 11111111] is:

- i) Class E
- ii) Class D
- iii) Class C
- iv) Class B
- i) IPv6 does not have a broadcast address:
 - i) True
 - ii) False
- j) In Transaction Oriented TCP, number of packets required:
 - i) 2 to 3
 - ii) 7 to 8
- 3. a) How is IP datagram formed?
- b) How is a message sent and subsequently received?
- c) How does Fast Retransmit mechanism work in TCP?
- d) How is agent advertisement utilized and by whom?
- e) Distinguish between IP in IP Encapsulation and Minimal Encapsulation.
- f) What does Flow Label in IPv6 header identify? Who sets it and can it be changed during routing?
- g) What is classful addressing? How can the first and last addresses in a block be found in IPv4 addressing?
- h) What is subnet mask? What is the subnet mask of 202.39.47.9? Is there subnet mask for Class E addresses? Why/why not?
- i) What is QoS object in new IPv6?
- 4. a) What are the types of unicast address in IPv6? Mention the respective fields of any one type.
- b) What does the Traffic class in IPv6 header identify? How does it help in routing?
- c) How does M-TCP protocol work?
- d) Explain reverse tunneling with the help of a suitable diagram.
- e) Who sends binding update to whom? Who maintains binding cache? What is binding acknowledgement?
- f) How does smooth handover differ from fast handover?
- g) What is DHCP lease?
- h) What is the subnet mask of 217.39.47.9?
- i) A block of address is assigned to an institution. One of the addresses is 205.16.37.39/28. What is the first address in the block?

Group C [CO3: Total Marks: 25 (=5x1+2x10)]

Question no. 5 is COMPULSORY [Marks: 05]

Answer ANY TWO Questions from No. 6, Question No. 7, Question No.8

5. Pick the correct answer:

- a) Response code 5XX in HTTP means:
 - i) successful
 - ii) redirection
 - iii) server error
 - iv) client error
- b) Adaptive Streaming is a hybrid of progressive download and streaming:
 - i) True
 - ii) False
- c) Real Time Streaming Protocol (RTSP) is dependent on transport protocol:

- i) False
- ii) True
- d) Major Entities of Session Initiation Protocol (SIP) include:
 - i) Proxy Server
 - ii) User agent
 - iii) Redirect Server
 - iv) all of the above
- e) Record types of DNS include A and AAAA:
 - i) False
 - ii) True
- 6. a) What do the following methods: PUT, PATCH, CONNECT perform?
- b) How is Round Trip Delay calculated in Real-Time Control Protocol (RTCP)?
- c) What is the role of the Resolver in DNS? What is iterative query in DNS?
- 7. a) What happens if header is missing in HTTP?
- b) What are the differences between GET and POST?
- c) Why are sequence nos. used in Real-Time Protocol (RTP) header? Mention any three payload types used in RTP.
- d) What are the fields in the query message header of DNS protocol?
- 8. a) What are the major components of DNS? What is recursive query in DNS?
- b) What does mdat denote?
- c) What happens if Header is missing in HTTP?
- d) How does a receiver using RTP find out packet loss?

Group D [CO4: Total Marks: 10]

Answer Question No. 9 OR Question No. 10 [Marks: 10]

- 9. Consider an NGO that accepts donations in cash or kind [clothes/any usable items]. Design and develop a module for registering a possible donor user.
- 10. Consider a digital platform that is used by students to buy old books and sell used books. Design and develop a module for registering a possible buyer/seller student.

Group E [CO5: Total Marks: 15]

Answer Question No. 10 OR Question No. 11 [Marks: 15]

- 10. a) In what type of network communication is Tunnel mode generally used?
- b) What is the difference in the authentication provided by AH and ESP in IPsec?
- c) How can sequence nos. be used to identify replay attack in IPsec?
- d) How is the server's certificate checked by the client?
- e) What are Change Cipher Spec protocol and Alert protocol?
- 11. a) How is message transmitted in Transport mode?
- b) Can AH and ESP be used together in IPsec? Why or Why not?
- c) What is security association in IPsec? How is it identified?
- d) How is sequence no. used in TLS?
- e) What are the different steps in the TLS Handshake protocol?

Sources:

- [3.2 Sem 2025.pdf](#)